

Git Changelog

Accelerated Video Processing System · Feb 23, 2026 – Mar 21, 2026

84	24	60	2	11
Total Commits	Merged PRs	Feature Commits	Contributors	Active Days

Contributors

Author	Commits	Merges	Regular
Abhinav Gupta	52	9	43
Pratik Chaudhari	32	15	17

Commit History

■ March 21, 2026

60ec5a5	Merge pull request #24 from rgabhi/fix_ab PR #24	Abhinav Gupta 13:46
add ppt and progress report		
ef43be5	add ppt and progress report	Abhinav Gupta 13:44
8b332d6	Merge pull request #23 from rgabhi/dev_prat PR #23	Pratik Chaudhari 11:30
Added version v1,v2		
b7e408f	Added version v1,v2	Pratik Chaudhari 11:30

■ March 20, 2026

cb1d3d5	Merge pull request #22 from rgabhi/dev_pratik PR #22	Pratik Chaudhari 23:12
Added DEMO Folder		
6b7e9c6	Added DEMO Folder	Pratik Chaudhari 23:11

4297c62	Merge pull request #21 from rgabhi/dev_pratik PR #21	Pratik Chaudhari 22:52
Fixed sepia		
df1ae01	Fixed sepia	Pratik Chaudhari 22:52
811a51a	Merge pull request #20 from rgabhi/dev_pratik PR #20	Pratik Chaudhari 16:59
Fixed emboss		
cafd3dc	Fixed emboss	Pratik Chaudhari 16:58
aea724c	Merge pull request #19 from rgabhi/fix_ab PR #19	Abhinav Gupta 14:24
sobel and gaussian optimizations		
17a1aa3	add readme	Abhinav Gupta 14:22
3d2e1cc	vectorize sobel	Abhinav Gupta 10:06
de60740	optimize gaussian -vectorize	Abhinav Gupta 09:35
a4944f8	optimize gaussian	Abhinav Gupta 09:17
e4ea94f	add back structural load	Abhinav Gupta 09:10
37d11ae	enhance sobel	Abhinav Gupta 08:45
c002f8b	enhance gaussianBlur	Abhinav Gupta 08:45

■ March 19, 2026

1c319c9	Merge pull request #18 from rgabhi/temp_dev PR #18	Abhinav Gupta 17:17
add verification and update main model		
9fb1612	update main file	Abhinav Gupta 16:20
304d692	add verification method	Abhinav Gupta 15:53
1cc4208	Merge pull request #17 from rgabhi/dev_pratik PR #17	Pratik Chaudhari 16:21
Update:MainViewModel		
32a766d	Update:MainViewModel	Pratik Chaudhari 16:20
85b9710	Merge pull request #16 from rgabhi/dev_pratik PR #16	Pratik Chaudhari 16:18
Added NativeLib		
36c8642	Added NativeLib	Pratik Chaudhari 16:17

cb8cee0	Merge pull request #15 from rgabhi/dev_pratik PR #15	Pratik Chaudhari 15:39
filters for Image		
ef3152b	filters for Image	Pratik Chaudhari 15:39

■ March 18, 2026

758b3c9	Merge pull request #14 from rgabhi/tmp_dev PR #14	Abhinav Gupta 23:33
add cpp base filter code		
e596e06	implement filters core	Abhinav Gupta 23:31
c258f2c	update build file	Abhinav Gupta 23:28
d4e1abb	Merge pull request #13 from rgabhi/dev_pratik PR #13	Pratik Chaudhari 23:09
Added:CMAKE_LIST,native_lib		
0905e89	Added:CMAKE_LIST,native_lib	Pratik Chaudhari 23:09

■ March 17, 2026

1db1c2c	Merge pull request #12 from rgabhi/dev_ab PR #12	Abhinav Gupta 22:26
app restructure		
08ffca1	restructure from other implementation	Abhinav Gupta 19:15
76e06db	Clear existing code to prepare for importing new project	Abhinav Gupta 19:10

■ March 11, 2026

fa8987c	add snaphot	Abhinav Gupta 19:14
6e69ddf	add data_store	Abhinav Gupta 18:19
acf926e	add data_store	Abhinav Gupta 17:55

■ March 10, 2026

7c0082e	update cam preview	Abhinav Gupta 03:13
44f46a8	update App skeleton and preview	Abhinav Gupta 03:02
5e01f0a	add mode selector and log view	Abhinav Gupta 02:31

■ March 09, 2026

1f306ff	add spark lines	Abhinav Gupta 12:01
cded2d1	add metrics tiles	Abhinav Gupta 11:57
b36f447	add main view model	Abhinav Gupta 11:51
120090c	fix button	Abhinav Gupta 11:26
360fd91	fix tilt inversion	Abhinav Gupta 11:17
a0ecbf0	add inversion op	Abhinav Gupta 11:12
9a7594d	add inversion op	Abhinav Gupta 11:04
f099a6c	add process frame fun	Abhinav Gupta 10:32
d57c8f7	add live feed	Abhinav Gupta 10:18
92fe57b	add permission button	Abhinav Gupta 09:56
3c50fe3	add cam permission	Abhinav Gupta 09:55
f6e0727	basic setup	Abhinav Gupta 09:48

■ February 27, 2026

7c7154f	Delete app/src/main/drawable directory	Pratik Chaudhari 13:13
95f1908	Merge pull request #11 from rgabhi/dev_pratik PR #11	Pratik Chaudhari 13:11
Fixed UI for filter,switch mode		
626a6c2	Fixed UI for filter,switch mode	Pratik Chaudhari 13:10
c559cb7	Merge pull request #10 from rgabhi/dev_pratik PR #10	Pratik Chaudhari 12:54
Fxiied camera error		
6f8f077	Fxiied camera error	Pratik Chaudhari 12:53
a1533fe	Merge pull request #9 from rgabhi/dev_pratik PR #9	Pratik Chaudhari 11:50
Added:Permission ,Thermal,Updated:libraries		
da1fc53	Added:Permission ,Thermal,Updated:libraries	Pratik Chaudhari 11:49
334e1c3	Merge pull request #8 from rgabhi/dev_abhi PR #8	Abhinav Gupta 11:35
Dev abhi		
b959037	add res layout configs	Abhinav Gupta 11:34
27f977a	add add dashboard_bg configs	Abhinav Gupta 11:30

71e64aa	add cpu load reading	Abhinav Gupta 11:25
e92ea2a	add dashboard update method	Abhinav Gupta 11:24
9cdd56b	Merge pull request #7 from rgabhi/dev_pratik PR #7	Pratik Chaudhari 10:42
Added:Thermal monitor ,Performance Tracker ,Baseline Processor		
7ee20bf	Added:Thermal monitor ,Performance Tracker ,Baseline Processor	Pratik Chaudhari 10:41
c5cab58	Merge pull request #6 from rgabhi/dev_abhi PR #6	Abhinav Gupta 10:29
Dev abhi		
da6489c	add frame file	Abhinav Gupta 10:08
d069ebf	add filter file	Abhinav Gupta 10:07
68270c1	add exec mode file	Abhinav Gupta 10:03
675e5b6	Merge pull request #5 from rgabhi/dev_pratik PR #5	Pratik Chaudhari 01:33
added frame processing pipeline		
9cfa03e	added frame processing pipeline	Pratik Chaudhari 01:32
52d732b	Merge pull request #4 from rgabhi/dev_abhi PR #4	Abhinav Gupta 00:06
Dev abhi		
39a7a43	add yuv converter	Abhinav Gupta 00:04
ba45ece	camera setup	Abhinav Gupta 00:00

■ February 24, 2026

5909059	Merge pull request #3 from rgabhi/dev_pratik PR #3	Pratik Chaudhari 23:12
Added empty MainActivity		
babc120	Added empty MainActivity	Pratik Chaudhari 23:12
7160fe9	Merge pull request #2 from rgabhi/dev_abhi PR #2	Abhinav Gupta 22:48
Dev abhi		
3e588fe	setup config	Abhinav Gupta 22:45
3a8eee8	setup config	Abhinav Gupta 22:45

b3ddd1 4	Merge pull request #1 from rgabhi/dev_pratik PR #1	Pratik Chaudhari 10:25
initialize android manifest with permission		
e67249 0	initialize android manifest with permission	Pratik Chaudhari 10:24

■ February 23, 2026

fd40dec	Initial project setup with Gradle configuration	Pratik Chaudhari 23:10
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Phase 2: ARM NEON Acceleration

Day 21 – 40 ✓
Complete

Phase 2 was the most technically intensive phase. What began as a straightforward port of Kotlin filters to C++ NEON intrinsics evolved into a multi-round performance engineering exercise driven by profiling and diagnosis of each round's bottlenecks.

7.1 The Optimization Journey

Stage	FPS	Description
Naïve SIMD	~5 ↓	Gather/scatter anti-pattern — slower than baseline
Round 1	~12	Separable filters + two-pass decomposition
Round 2	20–30	Structural loads + fixed-point arithmetic
Round 3	24–31 ✓	Overlapping loads + pure bitwise kernel math

7.2 The Gather/Scatter Anti-Pattern

The initial SIMD implementation was **slower than the Kotlin baseline**. The root cause: manually packing pixel data byte-by-byte into arrays before loading them into NEON registers — the gather/scatter anti-pattern.

■ The Fundamental Rule of SIMD

Data must be **contiguous in memory**. If the CPU must search around memory to collect individual bytes before vector math, the gathering overhead destroys any parallelism benefit.

7.3 Key Optimization Strategies

7.3.1 Separable Gaussian Blur

The 2D 5×5 Gaussian kernel is **linearly separable** into two 1D passes, reducing multiply-adds from 25 → 10 per pixel per channel — a **60 % reduction**. Kernel weights {1, 4, 6, 4, 1} sum to 16 (power of 2), enabling **pure bitwise arithmetic** (zero multiplications):

```
uint16x8_t sum = vmovl_u8(p_m2.val[c]);

sum = vaddq_u16(sum, vshlq_n_u16(vmovl_u8(p_m1.val[c]), 2)); // ×4

uint16x8_t center = vmovl_u8(p_0.val[c]);

sum = vaddq_u16(sum, vaddq_u16(vshlq_n_u16(center, 2),
vshlq_n_u16(center, 1))); // ×6

out.val[c] = vshrn_u16(sum, 4); // ÷16 = >>4
```

7.3.2 Fixed-Point Grayscale (Sobel Pass 1)

Floating-point conversions are expensive on mobile. Fixed-point arithmetic eliminates them entirely, processing **16 pixels per iteration** via `vld4q_u8` structural loads:

$$Gray_{float} = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B$$

■

`Grayint = (77·R + 150·G + 29·B) >> 8`

÷256 becomes a single-cycle bit-shift (>> 8)

7.3.3 Vectorized Sobel (Pass 2)

Overlapping loads exploit L1 cache on the 1-byte-per-pixel grayscale buffer, computing gradient magnitude via a single hardware instruction:

```
// |a - b| in a single hardware instruction
uint16x8_t abs_gx = vabdq_u16(col_right, col_left);
uint16x8_t abs_gy = vabdq_u16(row_bottom, row_top);

// Saturating narrow: clamps to 255 automatically
uint8x8_t mag = vqmovn_u16(vaddq_u16(abs_gx, abs_gy));
```

7.4 Performance Results

Round	Gaussian (FPS)	Sobel (FPS)	Key Technique
Baseline (Kotlin)	~10	~15	Scalar for-loops
Naïve SIMD	~5 ↓	~10 ↓	Gather/scatter anti-pattern
Round 1	~12	~20	Separable + two-pass
Round 2	—	~30 ✓	Structural loads + fixed-point
Round 3 (Final)	~24 ✓	~31 ✓	Overlapping loads + bitwise math

7.5 Understanding the Performance Ceilings

Sobel @ 30 FPS — Camera Sensor Limit	Gaussian @ 24 FPS — The Memory Wall
Sobel processes each frame in ~10 ms, but CameraX delivers frames at 30 FPS (33.3 ms intervals). The CPU outpaces the camera sensor. Breaking past 30 FPS requires Camera2 Interop configured for 60 FPS mode.	Separable Gaussian requires an 8 MB temporary ARGB buffer for 1080p frames — exceeding the L1/L2 cache. SIMD instructions execute in sub-nanoseconds, but the CPU waits milliseconds for data from main RAM. Solutions: cache-blocking (tiling) or GPU offloading.

Filter Specifications & SIMD Strategies

Filter	Kernel	Workload	SIMD Strategy
Sepia	1×1	Light arithmetic (BT.601 + bias)	<code>vmulq_f32 + vmlaq_f32</code> , 4 px

Gaussian Blur	5×5	Heavy spatial conv., memory-BW bound	Separable 1D passes + vshlq
Sobel Edge	3×3	Two-pass: grayscale → gradient mag.	vld4q_u8 + vabdq_u16
Emboss	3×3	Directional conv. + bias 128	vmlaq_s32 multiply-accumulate
Vignette	1×1	Per-pixel center distance, math-heavy	vrsqrteq_f32 hw reciprocal $\sqrt{}$

✓ Phase 2 complete — ARM NEON SIMD acceleration fully implemented for all 5 filters.